

**Employer Side
- ATS
- AI hiring agents**

Impact of AI-exposure on Labor Demand: Firm-Size Heterogeneity

Minjae Seo

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Abstract

The November 2022 release of ChatGPT marks the first general-purpose deployment of a large language model capable of substituting for cognitive labor at scale. Whether this technological shock reshaped firms' hiring decisions, and whether the response was symmetric across the firm-size distribution, are open empirical questions whose answers have first-order implications for the future of work, market structure, and the diffusion of general-purpose technologies. This proposal lays out the design for a triple-difference (DDD) study that exploits three margins of variation: occupation-level LLM exposure (Eloundou et al. 2023), within-industry firm size, and the pre/post-ChatGPT split, applied to a 14.5-million-cell panel of U.S. public-firm job postings from Revelio Labs merged to Compustat fundamentals, 2018-2025. Preliminary estimates show that within-size $\text{AIExp} \times \text{Post}$ coefficients are sharply negative at both small (-0.211 , SE 0.033) and large (-0.146 , SE 0.026) firms, but the triple interaction itself is not statistically distinguishable from zero ($+0.065$, $p = 0.14$, 95% CI $[-0.021, +0.151]$). I read this as suggestive but not yet conclusive evidence against the Brynjolfsson-Bloom prior that large firms lead in IT adoption: the point estimate runs the other way but inference is too noisy to rule out a null. The proposed work will (i) extend the outcome from postings to separation rates and WARN-Act layoffs, where the candidate channels predict a sharper sign reversal; (ii) revisit the design with PPML estimation as a robustness on the $\log(\cdot + 1)$ specification; and (iii) test against alternative exposure measures (Felten AIOE) and treatment dates (GPT-3, Copilot, GPT-4).

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1 Introduction and Research Question

1.1 Motivation

The November 2022 release of ChatGPT crossed a deployment threshold no prior labor-saving technology had crossed within a comparable window: a general-purpose model with measurable substitutability for cognitive tasks reached one hundred million users in two months. The 2023-24 cycle of white-collar layoffs at Meta, Amazon, Google, Microsoft, and Salesforce (roughly half a million positions) ran in parallel with the diffusion of generative AI tools, but the data confound two distinct stories: substitution of cognitive tasks by LLMs, and a delayed correction to the 2021 post-COVID hiring boom. Disentangling them requires variation that the aggregate time-series cannot supply.

The existing AI-exposure literature scores *occupations*. Felten, Raj, and Seamans (2021) construct an Artificial Intelligence Occupational Exposure index from mTurk ratings of how ten AI capabilities map to O*NET work activities; Eloundou et al. (2023) ask GPT-4 to evaluate, task by task, whether a large language model with software tooling can cut completion time by 50 percent or more. Both indices treat exposure as a property of the occupation, not the employer. They are informative about *which workers* face a substitutability risk but silent about *which firms* responded. A natural first cut on the firm-side heterogeneity is firm size: large firms have capital depth, internal labor markets, and shareholder pressure; small firms face cash constraints, no procurement delay, and no political cost to visible workforce reductions. Brynjolfsson and Hitt (2000) and Bloom, Sadun, and Van Reenen (2012) document that large firms historically led in adopting general-purpose IT, suggesting that large firms should also lead in the labor-demand response to generative AI.

Indeed Hiring Lab data through November 2025 (Figure 1) make the firm-size dimension visible at the aggregate. AI-related postings have tripled since 2018 but remain rare overall (about 6 percent of U.S. firms post any AI-related role by 2025), and the concentration is striking: ninety percent of all AI postings come from one percent of employers. Top-one-percent firms exhibit 49.9 percent AI-posting adoption; the bottom third, 1.3 percent. Small firms grew faster in *rates* (+318 percent bottom-third vs. +31 percent top-third) but from a low base, and the absolute gap widened.

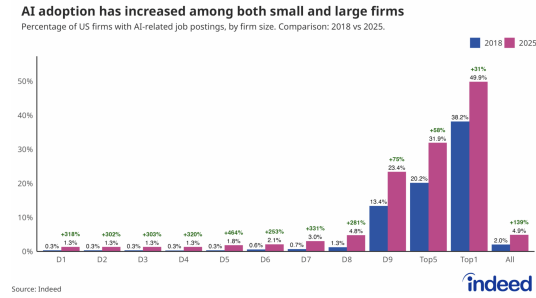
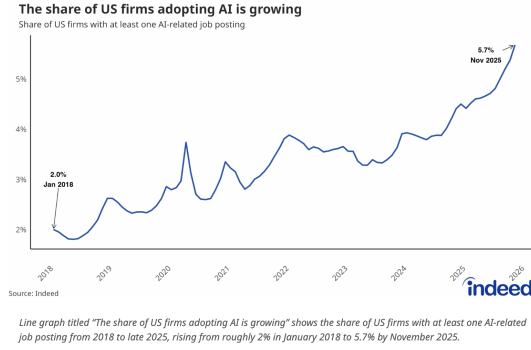


Figure 1: AI adoption is concentrated among the largest firms. Source: Gallacher, “AI Adoption Is Accelerating but Still Concentrated Among the Largest Firms,” Indeed Hiring Lab, January 16, 2026. Firm size is proxied by total posting volume on Indeed; an AI adopter is defined as a firm with at least one AI-related posting in the period.

1.2 Research question

After the November 2022 ChatGPT release, did labor demand for AI-exposed occupations contract more sharply at small firms than at large firms, relative to the pre-existing within-firm gap between AI-exposed and non-exposed roles?

The question is structured around three margins of identifying variation: (i) occupation-level LLM exposure (continuous Eloundou β score on $[0, 1]$), (ii) firm size (top employment quartile within 2-digit NAICS and fiscal year), and (iii) calendar time (before vs. after November 2022). Identification rests on a triple-difference (DDD) design that absorbs each two-way margin: firm-occupation matching, occupation-time aggregate shocks (rate cycle, sector decline), and firm-time aggregate shocks (post-COVID overhiring correction, firm-specific revenue path).

1.3 Contribution

This proposal contributes to two literatures.

Technology and labor demand. The empirical AI-labor literature has established that *occupations* exposed to AI have experienced measurable demand shifts (Acemoglu, Autor, Hazell, and Restrepo 2022; Babina, Fedyk, He, and Hodson 2024), but the within-firm evidence is limited. By holding the firm fixed and exploiting variation in within-firm occupational exposure, the design isolates the AI response from firm-wide cycles, including the post-COVID hiring correction and the 2022-23 rate tightening that hit cash-flow-sensitive sectors disproportionately.

Firm size and technology diffusion. The classical Brynjolfsson-Bloom prior that large firms lead in IT adoption is the empirical benchmark. The preliminary DDD point estimate on the labor-demand margin runs the other way: large firms appear to *cushion* the AI exposure penalty, though the cross-size difference is not statistically separable at conventional levels. I read this not as a contradiction of the prior but as evidence that the *outcome* matters. Postings, employment, and separations measure different margins of the same adjustment, and large firms have access to adjustment channels (internal reallocation, attrition) that small firms do not.

2 Data and Sample Construction

The empirical design rests on three data sources, one for each margin of identifying variation. Revelio Labs supplies the labor-demand outcome, Compustat supplies the firm-size treatment and balance-sheet controls, and Eloundou et al. (2023) supplies the occupation-level LLM exposure index. Table 1 names each source and its role in the design.

Table 1: Data sources and their role in the design.

Source	Role in the design
Revelio Labs postings (WRDS)	Outcome variable. Posting counts and average advertised salary at the firm $\times$ occupation $\times$ month level.
Compustat fundamentals (WRDS)	Firm-size treatment and controls. Employment, sales, assets, market value, 2-digit NAICS industry, and balance-sheet variables (R&D intensity, leverage, Tobin’s Q).
Eloundou et al. (2023)	Treatment intensity. Occupation-level LLM exposure score $\beta \in [0, 1]$ based on GPT-4 evaluation of O*NET tasks.

2.1 Revelio Labs job postings

Revelio Labs is a labor-market data vendor that ingests U.S. online postings from LinkedIn, Indeed, and direct employer sources, and standardizes them for research use through its WRDS license. Each posting is tagged with a 6-digit O*NET occupation code, a posting month, and a firm identifier. The raw file covers several hundred million U.S. postings between 2015 and 2025.

For this paper I aggregate the postings to firm $\times$ O*NET-occupation $\times$ month cells, keeping the posting count (median 2, mean 9.8, maximum 146,606 in the largest cell) and the mean advertised salary. The posting count is the *labor-demand* signal that drives the analysis: if generative AI substitutes for the tasks of a given occupation, firms should post fewer roles in that occupation after November 2022. The dependent variable in every preliminary regression is $y_{ijt} = \log(\text{n_postings}_{ijt} + 1)$.

I want to flag the $\log(\cdot + 1)$ transformation as a known weak link in the specification. The distribution of posting counts is highly right-skewed (median 2, max above 146{,}000) and contains many zero-posting cells, which the $+1$ shift accommodates mechanically but not cleanly: in the recent measurement literature, the elasticity interpretation under $\log(y + 1)$ is sensitive to the units of y (Chen and Roth 2024), and the more principled alternative for nonnegative counts with many zeros is Poisson PML, which estimates an elasticity directly without the unit-dependence problem. I report $\log(\cdot + 1)$ as the baseline because the three-way fixed-effect specification runs in tractable time on my hardware, whereas PPML on the full panel with $\alpha_{ij} + \gamma_{jt} + \delta_{it}$ is substantially more demanding. Re-estimating the main DDD with PPML on the estimation sample is a priority robustness; if the within-size coefficients are not robust to PPML, the interpretation in Section 4 will need to be revised.

2.2 Compustat firm fundamentals

Compustat is the S&P Global standardized financial database for U.S. public firms; through WRDS I pull the annual fundamentals file. Compustat plays two distinct roles in the design.

First, it defines the *firm-size treatment*. Within each (2-digit NAICS, fiscal-year) cell, I rank firms by employment, assign quartiles, and set $\text{Large}_i = 1$ for firms in the top quartile. The within-industry construction is deliberate: it prevents Large_i from mechanically loading onto the tech sector and lets the cutoff drift with industry composition and the macro cycle.

Second, it supplies the *balance-sheet controls* that enter the channel diagnostics in Section 5: log sales, log assets, log market value, Tobin’s Q , leverage, and R&D intensity. These do not enter the baseline DDD, where firm-time fixed effects absorb everything observable at the firm-year level, but they are used in the cash-constraint and procurement-frictions extensions.

The Compustat universe restricts the analysis sample to U.S. public firms; 24 percent of Revelio rows have no Compustat counterpart and drop out of the regression sample. These are predominantly smaller, younger, or delisted firms, a survivorship concern discussed in Section 6.

2.3 Eloundou et al. (2023) LLM-exposure scores

The exposure measure comes from Eloundou et al. (2023), who ask GPT-4 to read each O*NET task and answer whether a large language model can reduce the time to complete that task by at least 50 percent. Task-level yes/no answers are aggregated to occupation level using O*NET task weights, yielding three scores in $[0, 1]$: α (LLM alone, no tools), β (LLM with software and tools), and ζ (combined broader index).

I use β as the primary measure of AI exposure because the November 2022 ChatGPT deployment shipped with tooling (Code Interpreter, browsing, plug-ins), which is exactly the deployment scenario β is constructed to capture. The α and ζ measures, as well as the older Felten AIOE index, are reserved for robustness. The score enters the regression *continuously* on $[0, 1]$; the tercile cut shown in Figure 2 is used only for the descriptive plots in Section 4. The distribution across the 923 O*NET occupations is bimodal, with a left mass of low-exposure manual and service occupations and a right mass of cognitive and clerical occupations clustered around $\beta \approx 0.5$.

One temporal-validity concern is worth flagging up front. The β score was constructed in 2023 by asking GPT-4 to evaluate O*NET tasks; the *measurement instrument itself* did not exist in the pre-period. Whether the score is a clean ex-ante measure of substitutability or partly reflects post-treatment knowledge of which occupations were already adjusting is not testable from the index alone. I treat the score as fixed at the occupation level and rely on the within-occupation pre-period variation (which contains no AI exposure differential because β_j is time-invariant) to absorb any level effect. The remaining concern is differential measurement error across the β distribution, which would attenuate the within-size coefficients toward zero by a classical-errors argument but does not have an obvious sign for the cross-size difference. A robustness using the Felten AIOE index, which was constructed in 2018-2021 from a different evaluation procedure entirely, is the natural

cross-check.

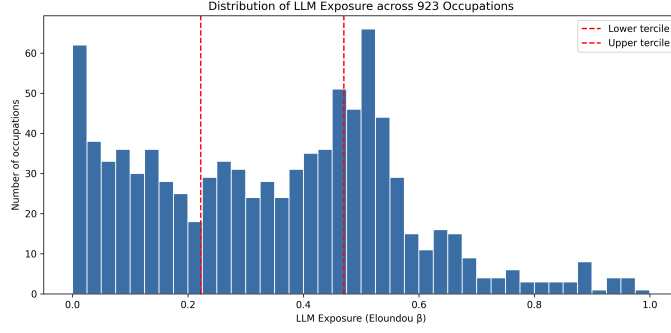


Figure 2: Distribution of Eloundou β across O*NET occupations. The distribution is bimodal: a left mass of low-exposure manual and service occupations ($\beta < 0.2$) and a right mass of cognitive and clerical occupations clustered around $\beta \approx 0.5$. Dashed lines mark the 33rd and 67th percentiles, used for descriptive plots only.

2.4 Sample construction and summary statistics

The merged firm $\times$ occupation $\times$ month panel covers January 2018 to December 2025 (96 months). For the regression sample I impose two restrictions: (a) cells must have a non-missing firm-size indicator (76.4 percent coverage; missing cases are firms outside the Compustat universe), and (b) the analysis window starts in January 2021 to avoid the pre-COVID and early-COVID structural break in posting behavior. Table 2 reports the resulting sample.

Table 2: Sample construction and summary statistics.

Quantity	Value
<i>Full panel (Jan 2018 to Dec 2025)</i>	
Firm $\times$ occupation $\times$ month cells	14,530,633
Distinct firms	10,363
Distinct occupations	917
Months covered	96
<i>Variable coverage</i>	
Coverage: AI exposure (β)	100.0%
Coverage: large-firm indicator	76.4%
<i>Pre / post split (November 2022)</i>	
Pre-ChatGPT cells	6,148,004
Post-ChatGPT cells	8,382,629
<i>Distributional moments</i>	
Mean Eloundou β	0.465
Posting count (median / mean / max)	2 / 9.8 / 146,606
<i>Estimation sample (post-2021, non-missing size)</i>	
Cells	9,810,059
Firms (clusters)	4,890
Occupations	917

3 Empirical Strategy

3.1 Specification

For firm i , occupation j , month t , with $y_{ijt} = \log(\text{n_postings}_{ijt} + 1)$, the baseline triple-difference is:

$$\begin{aligned} y_{ijt} = & \beta_1 \text{AIExp}_j \cdot \text{Post}_t + \beta_2 \text{Large}_i \cdot \text{Post}_t \\ & + \beta_3 \text{AIExp}_j \cdot \text{Large}_i \cdot \text{Post}_t \\ & + \alpha_{ij} + \gamma_{jt} + \delta_{it} + \varepsilon_{ijt}, \end{aligned} \quad (1)$$

where $\text{Post}_t = \mathbf{1}[t \geq \text{Nov 2022}]$. The three two-way fixed effects play distinct roles, summarized in Table 3.

Table 3: Role of each fixed effect in the DDD specification.

Fixed effect	Absorbs	Identifying concern addressed
α_{ij} (firm $\times$ occupation)	Time-invariant firm-occupation matching	Software firms post programmers; banks post analysts
γ_{jt} (occupation $\times$ time)	Aggregate occupation-specific shocks	Fed tightening drives down cognitive-occupation demand everywhere
δ_{it} (firm $\times$ time)	Firm-wide shocks at any horizon	Post-COVID overhiring correction; firm revenue path; sector cycle

Standard errors are clustered at the firm level.

3.2 Coefficient interpretation

A clarification is required on the role of β_1 and β_2 . Under the saturated fixed-effect structure, $\text{AIExp}_j \cdot \text{Post}_t$ is collinear with γ_{jt} (which contains a separate dummy for every occupation-month) and so β_1 is not separately identified once γ_{jt} is included. By the same argument, $\text{Large}_i \cdot \text{Post}_t$ is collinear with δ_{it} once δ_{it} contains a full set of firm-month dummies, since Large_i is firm-invariant within a year.

In practice, pyfixest reports a coefficient on $\text{Large} \times \text{Post}$ in column (1) of Table 4 because Large_i in my construction is *industry-year* relative — the top-employment-quartile assignment is reset each fiscal year — so Large_i does carry within-firm time variation when a firm crosses the quartile boundary between years. That residual variation is what β_2 in Table 4 is identified off. The number should be read as a within-firm switching effect on quartile transitions, not as a pure between-firm comparison; I do not put weight on it in the discussion. Only β_3 is the object of interest. Its population analog is the cross-size difference of the within-firm, across-occupation pre/post second difference,

$$\beta_3 = E[\Delta\Delta y \mid \text{Large}] - E[\Delta\Delta y \mid \text{Small}],$$

where $\Delta\Delta y \equiv (y_{\text{HIGH},\text{post}} - y_{\text{LOW},\text{post}}) - (y_{\text{HIGH},\text{pre}} - y_{\text{LOW},\text{pre}})$ for HIGH-vs-LOW exposure occupations. The key analytical property is that *any symmetric within-firm headcount shock* (a uniform 10-percent cut across all roles) cancels in β_3 once δ_{it} is included. Generative AI, which strikes exposed occupations differentially within the firm, can generate a nonzero β_3 ; the post-COVID correction cannot.

3.3 Why triple-difference and not two-way DiD

Figure 3 plots the high-minus-low log-posting differential, three-month moving average, indexed to January 2022 = 0. An OLS fit of the differential on time over the pre-period (January 2021 to October 2022) gives a slope of approximately -10 index points per year. The differential was deteriorating well before ChatGPT.

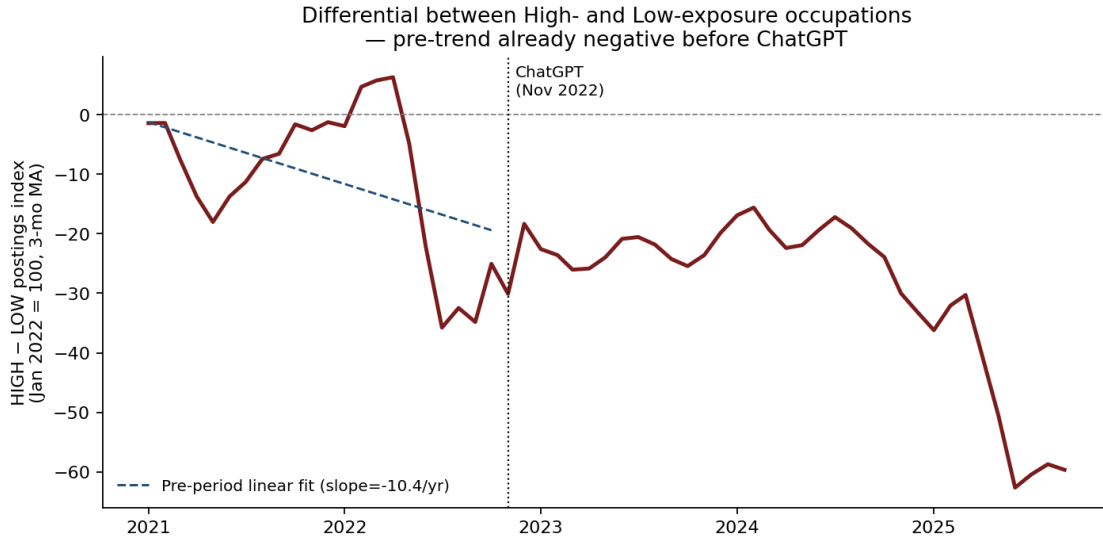


Figure 3: The HIGH-LOW exposure differential trends downward before ChatGPT. A two-way DiD attributes the post-period change to treatment alone; the pre-trend in the differential biases that estimate. The DDD differences out the common path and identifies the AI effect off the cross-size deviation from the trend.

The mechanical bias of a two-way DiD is $\hat{\beta}^{\text{DiD}} = \beta^{\text{true}} + (\widehat{dg_t/dt}) \cdot \bar{t}_{\text{post}}$. With a pre-trend slope of -10 points per year and a post window of roughly 2.5 years, the bias is on the order of -25 index points, the same order of magnitude as the post-period treatment effect itself. The DDD identifying assumption is strictly weaker:

$$\underbrace{E[g_t \mid \text{Large}, t \leq T^*] - E[g_t \mid \text{Small}, t \leq T^*]}_{\text{cross-size difference of the differential}} \text{ is constant in } t \text{ for } t \leq T^*.$$

Any common shock that moves the HIGH-vs-LOW differential at all firms, such as the rate cycle, the hiring boom and bust, or sector-level cognitive demand, cancels in the cross-size difference. The pre-trend is what *motivates* the design, not what discredits it.

Figure 4 shows the analogous event-study coefficients on the HIGH-vs-LOW differential, plotted in three-month bins with reference bin $[-3, -1]$. The pre-period leads are not flat: they trend upward in $k = -12, -9$, sit near zero in $k = -6, -3$, and the post-period coefficients stabilize near -0.045 .

A caveat. Strictly speaking, Figure 4 is the event study on the within-firm HIGH-vs-LOW gap (pooled across firm size). The diagnostic that actually corresponds to the DDD identifying assumption is the event study on the *cross-size difference* of that gap, i.e., the coefficient on $\text{HIGH}_j \cdot \text{Large}_i \cdot \mathbf{1}[t \in \text{bin}_k]$. I have not yet produced that plot; the implementation requires careful handling of the three-way fixed effects in the leads, which I have not fully worked out. What Figure 4 *does* deliver is a necessary condition for DDD parallel trends to hold: the cross-size difference of pre-period leads can be flat only if neither size group’s pre-period leads explode on their own. The pooled pre-period coefficients in Figure 4 are bounded (roughly ± 0.05 log points) and trend back to zero by $k = -3$, which rules out the worst case in which one size group’s HIGH-LOW gap is diverging sharply before treatment. This is partial assurance, not the full test; the cross-size event-study figure is the next item on my list.

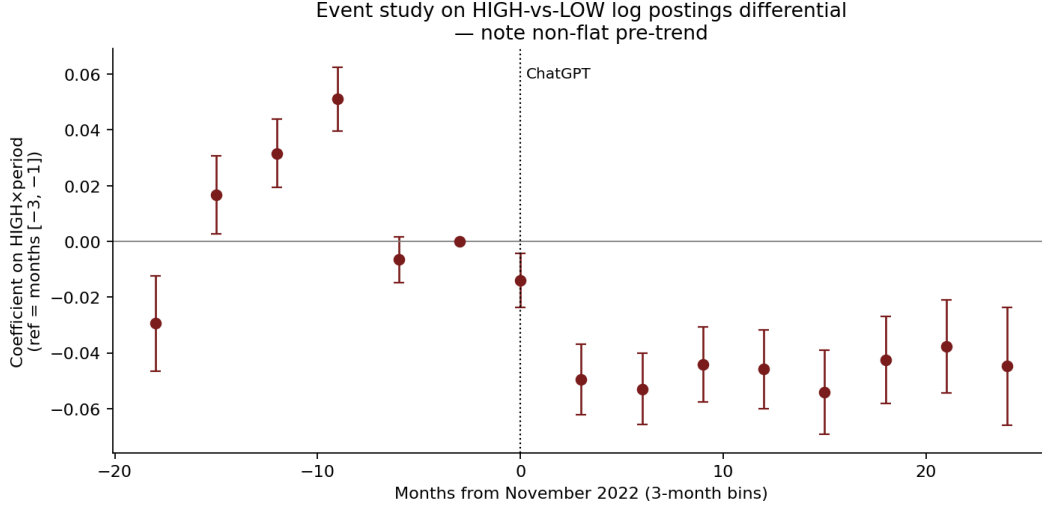


Figure 4: Event-study coefficients on the HIGH-vs-LOW log-posting differential. The pre-treatment differential is not flat; this is the diagnostic that motivates the DDD over a two-way DiD. The cross-size analog of this event study, identified off the DDD interaction, drops the common pre-trend out.

4 Preliminary Results

4.1 Descriptive: postings by exposure tercile

Figure 5 plots total U.S. postings indexed to January 2022 = 100, three-month moving average, split by Eloundou- β tercile (cut at the 33rd and 67th percentiles of the occupation distribution).

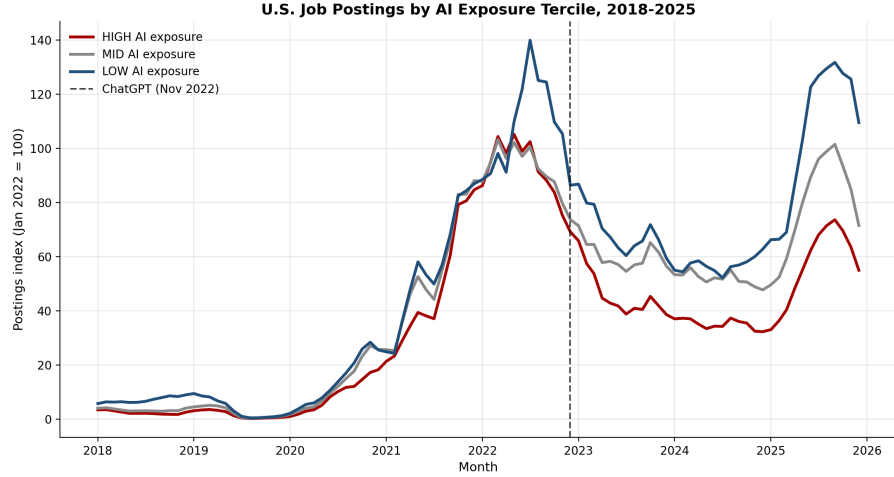


Figure 5: Postings by AI-exposure tercile, 2018-2025. HIGH-exposure occupations peaked lower in the 2021 boom and contracted fastest after November 2022. By 2025 the HIGH series sits 30 to 40 index points below LOW. The naive read that ChatGPT caused the gap is confounded by the 2021-2022 hiring boom and the rate cycle. The regression in Table 4 controls for both.

4.2 Descriptive: split by firm size

Figure 6 reproduces the tercile plot separately for large and small firms. Visual inspection suggests the post-2022 HIGH-LOW gap is *wider* at large firms, which was the original motivation for the question and the prior the design was set up to confirm. The regression flips the sign.

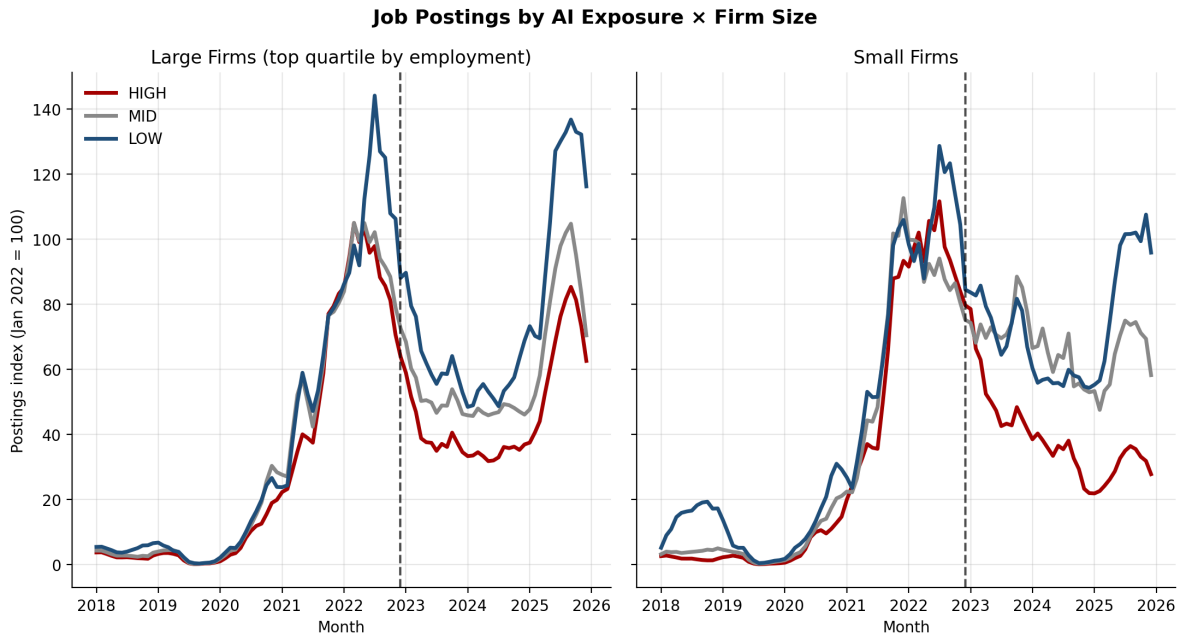


Figure 6: Postings by AI-exposure tercile, split by firm size. Large = top quartile of employment within (2-digit NAICS, fiscal year). Visual reading: the post-2022 HIGH-LOW gap looks wider at large firms. The DDD regression, which differences out firm-wide overhiring corrections and occupation-wide rate effects, reverses this reading.

4.3 Regression: triple-difference and split samples

Table 4 reports the baseline DDD (column 1) and the split-sample regressions that recover the within-size AI-exposure coefficient (columns 2 and 3). The point estimates run opposite to the descriptive reading.

Table 4: Triple-difference estimates of the post-ChatGPT change in log postings.

	(1) Pooled DDD	(2) Small firms	(3) Large firms
AIExp $\times$ Post	−0.211*** (0.033)	−0.211*** (0.033)	−0.146*** (0.026)
Large $\times$ Post	−0.057** (0.025)		
AIExp $\times$ Large $\times$ Post	+0.065 (0.044)		
Firm $\times$ occupation FE	✓	✓	✓
Occupation $\times$ time FE	✓	✓	✓
Firm $\times$ time FE	✓	✓	✓
Observations	9,810,059	3,159,705	6,650,354
Firms (clusters)	4,890	3,910	1,265

Dependent variable: $\log(n_postings_{ijt} + 1)$. Sample restricted to cells from January 2021 onward with non-missing firm-size indicator. Eloundou β enters continuously on $[0, 1]$. Large_{*i*} = 1 for firms in the top employment quartile within (2-digit NAICS, fiscal year). Post_{*t*} = $\mathbf{1}[t \geq \text{Nov 2022}]$. Standard errors clustered at the firm level in parentheses. Two-way significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Three things to read off the table. First, the within-size AIExp $\times$ Post coefficients are individually sharp: −0.211 at small firms and −0.146 at large firms, both at the 1 percent level. Second, the triple interaction is positive (+0.065) but with a standard error of 0.044 and a 95 percent confidence interval of $[-0.021, +0.151]$. It does not separate from zero at conventional levels ($p = 0.14$).

The honest summary is therefore: *the point estimate is consistent with large firms cushioning the AI-exposure penalty on the posting margin, but I cannot reject the null of no cross-size difference on this outcome*. This is weaker than the Brynjolfsson-Bloom-style counterfactual would predict (if large firms led in adopting AI tools, the triple interaction should be sharply *negative*, not weakly positive). Calling this a clean reversal of the prior overstates what the data say at $N \approx 10^7$ with firm-clustered standard errors.

In my view, the most informative way to read these numbers is as a constraint on the next stage of the design rather than as a finished result. The posting outcome is a noisy proxy for the AI-driven labor adjustment at large firms because of the internal-labor-market and visibility channels discussed in Section 5. If those channels are real, the cross-size asymmetry should be sharper on the separation and WARN-Act outcomes. That is where the design earns or fails to earn the “reversal” claim, not in Table 4.

5 Channels and Interpretation

The cross-size asymmetry on the posting outcome admits several mechanisms, but two of them carry most of the interpretive weight. The first is *internal labor markets*: large firms have somewhere to put exposed workers and small firms do not. A bank with three hundred risk analysts and a thirty-person AI tooling group can quietly move people across functions when the marginal product of a risk analyst falls; a twelve-person startup cannot. If this channel binds, the AI shock shows up at large firms not as a cut in postings but as a within-firm shift in the *internal* mix of roles, with external posting growth absorbed into the firm-wide attrition rate. The natural test re-aggregates the Revelio individual-positions module to the worker level for a matched subsample, codes role transitions inside the same firm, and asks whether the within-firm probability that an AI-exposed worker moves to a less-exposed role rises post-November 2022 differentially at large firms; this is a worker-level test of what the firm-level DDD sees only indirectly, and is planned for months 4-6.

The second main channel operates one layer up. Postings measure gross hiring intent, not net employment or separations: a large firm that reduces AI-exposed headcount by five percent through attrition over twelve months and never posts to backfill produces a *zero* posting response in my data, while a small firm with the same target cut has to post fewer roles because it has no attrition buffer at that scale. This makes the posting outcome a *biased* proxy for labor adjustment at large firms, and it is the reason re-estimating the DDD with $\text{sep}_{ijt} = \text{outflow}_{ijt} / \text{count}_{ij,t-1}$ from the Revelio workforce-dynamics module is the single highest-priority extension. If the channel is real, the triple-interaction coefficient should *change sign* on the separation outcome rather than merely shrink; that sign change, if it appears, is what would let me claim that the Brynjolfsson-Bloom prior fails on the labor-adjustment margin. Three further channels matter but are second-order in the design: procurement and security review at large firms (testable by dating the first 10-K LLM mention per firm and comparing DDD estimates before and after), cash-constraint heterogeneity in the 2022-23 rate cycle (testable by interacting Compustat liquidity ratios with $\text{AIExp} \times \text{Post}$), and the visibility and political cost of headcount moves at large firms (testable by comparing the WARN-Act involuntary-layoff outcome against the pooled-outflow outcome). The ranking above reflects both my read of the channel mechanics and the order in which the available Revelio modules let me probe each one; the remaining three are second-order in the design rather than in importance, and the goal in the next stage is to bring all five into a single channel table rather than set the smaller ones aside.

6 Limitations and Identification Threats

I list the threats in roughly the order I think they matter for the inferences in Section 4, not in textbook order.

Survivorship in the Compustat link is more serious than I initially treated it. Twenty-four percent of Revelio rows fail the merge into Compustat. Those rows are not missing at random: they are predominantly smaller firms, younger firms, and firms that delisted before the post-period ends. The selection operates exactly along the dimension that defines the Large_i treatment, which

means the “small firms” sample I estimate column (2) on is a *survivor* sample of small public firms, not a random sample of small employers. Concretely, if firms that disappeared from Compustat in 2023-25 cut AI-exposed postings more sharply (because they were failing), the surviving small-firm sample underestimates the true small-firm AI sensitivity, which makes the large-firm coefficient look closer to the small-firm coefficient than it really is. That biases the triple interaction *toward zero*, which is the direction my preliminary result points. I cannot discipline this without re-running the analysis on the universe of Revelio firms with size proxied from posting volume rather than Compustat employment; that robustness is on the priority list, but I have not yet done it. *In my view this is the single most serious threat to the headline result*, more than the pre-trend, more than the exposure-measure validity.

Outcome margin (postings are demand-side, not employment). The post-2022 large-firm posting series understates the AI-driven labor adjustment by the size of whatever can be done through attrition and internal mobility. Section 5 explains why I expect the cross-size asymmetry to be sharper on separation rates and WARN-Act layoffs. Until the separation-rate estimates are in hand, the posting result should be read as the *upper bound* on how attenuated the cross-size gap can be.

Exposure $\neq$ adoption. The Eloundou β score measures the *potential* for an LLM with tools to substitute for occupation tasks, not actual firm-level deployment. The treatment is therefore intent-to-treat with respect to AI adoption. A 10-K-based deployment-dating exercise (Section 5) is the most direct correction.

Sequential treatment. Generative AI did not arrive in a single shock. GPT-3 (June 2020), GitHub Copilot (June 2021), and GPT-4 (March 2023) form a sequence. I have not yet resolved how to handle the sequentiality cleanly: the November 2022 date matches the cultural and product-market moment but pools three distinct capability releases. Event studies around alternative dates are a planned robustness; if the post-period coefficients are mostly loading on March 2023 (GPT-4), the headline date is misleading.

Interest-rate confound. Rate hikes hit cash-flow-sensitive sectors more, and the labor-demand response in cognitive occupations maps onto AI-exposure imperfectly but not orthogonally. The firm-time fixed effect δ_{it} absorbs the firm-wide rate response. What it cannot absorb is the part of the rate response that varies *across occupations within a firm*. A direct interaction with sector-level funding-cost shocks is a planned channel diagnostic.

Specification (functional form). As discussed in Section 2.1, the $\log(\cdot + 1)$ transformation is a fragile choice for a count outcome with many zeros and an elasticity interpretation that depends on the units of the count (Chen and Roth 2024). PPML on the three-way-FE specification is the right robustness; I have it on the list but it is computationally heavy on $N \approx 10^7$ and I have not run it on the full sample yet.

7 Proposed Extensions and Timeline

The preliminary analysis establishes the panel and the design and documents a directional pattern of attenuation at large firms; whether that pattern is statistically separable from zero is what the next stage of work has to settle. The proposed dissertation extends the design along three axes, in roughly the order below. The timeline is approximate and assumes continued access to the Revelio modules I do not yet have pulled.

Outcome margin (priority 1; approximately months 1-4). The single most informative extension is to re-estimate the same DDD on the labor-flow margin. The separation rate $\text{sep}_{ijt} = \text{outflow}_{ijt} / \text{count}_{ij,t-1}$ derived from the Revelio workforce-dynamics module captures the within-firm attrition channel that postings cannot see; the WARN-Act layoff indicator from the Revelio layoffs module isolates involuntary separations and disambiguates them from voluntary turnover. If the internal-labor-markets and visibility channels in Section 5 are real, the triple-interaction coefficient should *change sign* (not merely shrink) on these outcomes.

Channel decomposition (priority 2; approximately months 4-8). The two main channels identified in Section 5 have direct tests that the outcome-margin extension above does not subsume. For internal labor markets, re-aggregate the individual-positions module to the worker level for a matched subsample of large and small firms, and test whether the within-firm probability of an exposed-to-unexposed role transition rises post-November 2022 differentially at large firms. For the three second-order channels, run the targeted tests sketched in Section 5: 10-K LLM-mention dating for procurement frictions, Compustat liquidity-ratio interactions for cash-constraint heterogeneity, and the WARN-Act-vs-pooled-outflow comparison for the visibility margin. The deliverable is a single channel table ranking the five mechanisms by their predictive content for the cross-size asymmetry.

Methodological robustness (priority 3; approximately months 6-10). Four robustness exercises that the headline result needs to survive: (a) re-estimate the main DDD with PPML instead of $\log(\cdot + 1)$, on the estimation sample first and on the full panel if computationally feasible; (b) replace Eloundou β with α , ζ , and the Felten AIOE index, the last of which was constructed pre-LLM and provides the cleanest measurement cross-check; (c) re-estimate around GPT-3 (June 2020), Copilot (June 2021), and GPT-4 (March 2023) as alternative treatment dates, to test how much the November 2022 result is loading on the GPT-4 release; (d) replace the firm-size definition with sales-based and market-cap-based within-industry quartiles, and report wild-cluster-bootstrap p -values at both the firm and firm- $\times$ -occupation cluster levels as the inferential robustness.

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